



Education

Lab 1. Setting Up the Course Lab Environment

Overview

Before we dive into the hands-on activities, let's configure the lab environment with the following tools:

- **Docker:** Used by kind to provision the Kubernetes cluster.
- **kind:** Used to create Kubernetes clusters.
- **kubecttl:** Used to manage the Kubernetes cluster.
- **Helm:** Used for Kubernetes package management.
- **Siege:** A load testing tool.

Exercise 1.1: Set Up the Course Lab Environment

Linux Environment Setup

While we present various setup options in this section, we will use an Ubuntu 24.04 host and a Kubernetes cluster setup using **kind**.

To set up **Docker**, follow these steps.

1. Install Docker:

```
curl -fsSL https://get.docker.com/ | sh
```

2. Enable Docker to start on boot:

```
sudo systemctl enable --now docker
```

3. Check that the Docker service is running:

```
sudo systemctl status docker
```

4. Add current user to Docker group:

```
sudo usermod -aG docker $USER
```

5. Refresh shell session (**by exiting & logging in again**).
6. Verify Docker installation:

```
docker ps
```

To install the latest version of **kubect**l, execute the commands below.

1. Download the binary:

```
curl -sSL -O "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
```

2. Modify permissions:

```
chmod +x kubectl
```

3. Move to **/usr/local/bin**:

```
sudo mv kubectl /usr/local/bin
```

To install the latest **Helm** version, perform the following steps.

1. Download Helm (if the link doesn't work, update the link with a version from [GitHub releases](#)):

```
curl -sSL -O https://get.helm.sh/helm-v3.19.0-linux-amd64.tar.gz
```

2. Extract the downloaded archive:

```
tar -zxf helm-v3.19.0-linux-amd64.tar.gz
```

3. Move the Helm binary to **/usr/local/bin**:

```
sudo mv linux-amd64/helm /usr/local/bin/helm
```

To install the latest version of **Siege**, execute the following command.

1. Install the binary:

```
sudo apt-get install siege
```

Siege is a multi-threaded HTTP/FTP load tester and benchmarking utility.

To install **kind**, run the following command.

1. For AMD64 / x86_64:

```
[ $(uname -m) = x86_64 ] && curl -Lo ./kind  
https://kind.sigs.k8s.io/dl/v0.30.0/kind-linux-amd64
```

2. For ARM64:

```
[ $(uname -m) = aarch64 ] && curl -Lo ./kind  
https://kind.sigs.k8s.io/dl/v0.30.0/kind-linux-arm64
```

3. Modify permissions:

```
chmod +x ./kind
```

4. Move the kind binary `/usr/local/bin`:

```
sudo mv ./kind /usr/local/bin/kind
```

5. Create cluster:

```
kind create cluster
```

Creating cluster "kind" ...

✓ Ensuring node image (kindest/node:v1.34.0) 

✓ Preparing nodes 

✓ Writing configuration 

✓ Starting control-plane 

✓ Installing CNI 

✓ Installing StorageClass 

Set kubectl context to "kind-kind"

You can now use your cluster with:

```
kubectl cluster-info --context kind-kind
```

Have a nice day! 

6. Verify cluster:

```
kubectl get ns
```

NAME	STATUS	AGE
default	Active	36s
kube-node-lease	Active	36s
kube-public	Active	36s
kube-system	Active	36s
local-path-storage	Active	30s

Metric Server Setup in Kubernetes

To install metrics server in Kubernetes, execute the commands below.

1. Add Helm repository:

```
helm repo add metrics-server https://kubernetes-sigs.github.io/metrics-server/
```

`metrics-server` has been added to your repositories.

2. Install metrics server:

```
helm upgrade --install metrics-server metrics-server/metrics-server -n
kube-system --set args[0]=--kubelet-insecure-tls
```

```
NAME: metrics-server
```

```
LAST DEPLOYED: Wed Oct 15 16:20:19 2025
```

```
NAMESPACE: kube-system
```

```
STATUS: deployed
```

```
REVISION: 1
```

```
TEST SUITE: None
```

```
NOTES:
```

```
*****
* Metrics Server *
*****
  Chart version: 3.13.0
  App version:   0.8.0
  Image tag:     registry.k8s.io/metrics-server/metrics-server:v0.8.0
*****
```

3. Verify installation:

```
kubectl get pods -n kube-system -l=app.kubernetes.io/name=metrics-server
```

NAME	READY	STATUS	RESTARTS	AGE
metrics-server-55fd7d4789-v6dpw	1/1	Running	0	63s

Congratulations, your environment is ready!